
Key Projects Overview

Jan. 2026

Overview List **AUTO**

Application	Product	Customer (OEM / End-user)	Seg.	Pot. [t/a]	SOP	Resp. (Sales/AD)
ADAS Cleaning Pump	Grivory HT1VA-35 HYS	Jihan (Seres/Valeo)	240	32	Q2.2026	SUNL / HEBA
eWGA Gear	Grivory HTV-5H1 & HT2V-3H LF	UAES (BYD&GWM)	270	96	Q3.2027	WAHU / HEBA
Roof Rack Base	Grivory GV XE 16101 Black 9839 (GVL-5H HP)	Kemflo (RIVIAN)	230	21	Q2 2026	ZHER / FUVI
Lamellar Bar	Grinova S XE 10767 Black L11720.4 (XGF35)	Chengdu AVIC (Dongfeng)	230	19	Q3 2026	XURO / ZHZA
Hard Floor Panel under Console	Grivory GVX-5H Black 9915	KBI Dongkook Ind. (HKMC)	230	60	Q1 2029	PAIS / SEOJ

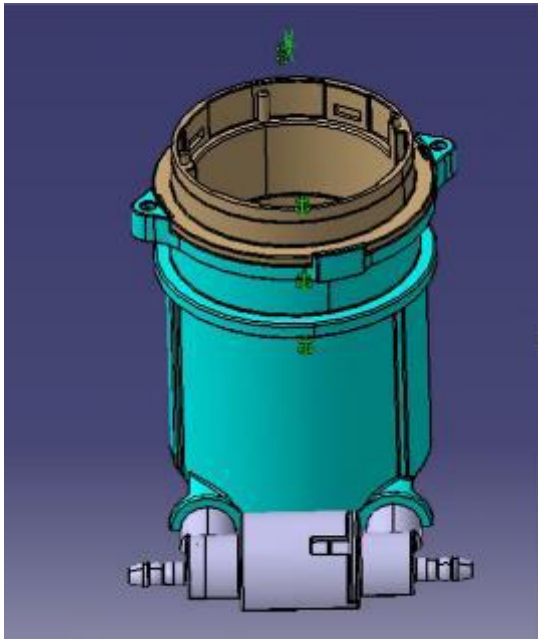
ADAS Cleaning Pump / HEBA



Material:
Grivory HT1VA-35 HYS Black 9225

Requirements

- Coolant resistance
- High toughness to solve crack issue
- Working pressure 0.8 MPa
- -40°C ~ 120 °C



Customer: Shanghai Jihan

OEM / End User: Seres

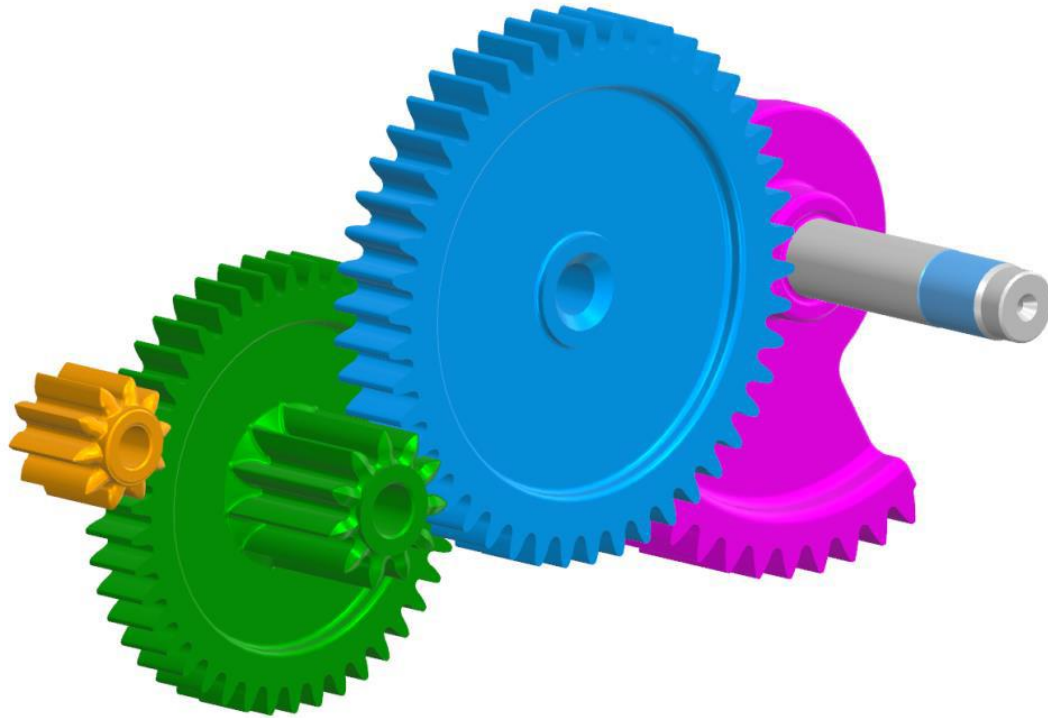
AD-TCS Key Project

ADAS Cleaning Pump / HEBA

Application	Material	Customer (OEM, End-user)	Project No. or Segment	Pot. [t/a]	SOP	Resp. [Sales/AD]
ADAS Cleaning Pump	Grivory HT1VA-35 HYS	Jihan (Seres/Valeo)	240-251212-2	32	Q2.2026	SUNL / HEBA
Advantages <i>(Technical Competition)</i>			Risks / Competitor Grade			
+ Better ultrasonic welding performance compared with PPS. + Good coolant resistance			- Customer got a lower quotation of PPA from local competitor			
Actual Situation			Next Steps <i>(by When)</i>			Resp.
■ Soft tooling was made to evaluate assembly ■ PA6T/X GF35 has replaced PPS GF40 in drawing			■ Suzhou Pollmann is building a mold for mass production (02.2026)			SUNL

AD-TCS Key Project

eWGA Gear / HEBA



Material:

Grivory HTV-5H1 Black 9205

Grivory HT2V-3H LF Black 9205

Requirements

- Heat aging resistance at 160 °C
- Gear strength min. 650N
- -40°C ~ 160 °C

Customer: UAES

OEM / End User: BYD & GWM

AD-TCS Key Project

eWGA Gear / HEBA

Application	Material	Customer (OEM, End-user)	Project No. or Segment	Pot. [t/a]	SOP	Resp. [Sales/AD]
eWGA Gear	Grivory HTV-5H1 & HT2V-3H LF	UAES (BYD&GWM)	270-251014-1	96	Q3.2027	WAHU / HEBA
Advantages <i>(Technical Competition)</i>			Risks / Competitor Grade			
+ Lower water absorption and denstiy, compare with PA46			- Stanyl TW271F6 (PA46+PTFE+GF30) - Stanyl TW278F10 (PA46+PTFE+GF50) - Stanyl TW241F12 (PA46+GF60)			
Actual Situation			Next Steps <i>(by When)</i>			Resp.
■ HT2V-3H LF was molded trial in the tooling of TW278F10 at molder. Dimension is over limit due to different GF content, but gear strength is better than PA46			■ Supply comprehensive PA46 replacement solution to UAES (01.2026)			HEBA

AD-TCS Key Project

Roof Rack Base / ZHER

Material:
Grivory GV XE 16101 Black 9839
(GVL-5H HP)

Requirements

- High stiffness and strength at elevated temperature
- 500 hrs weathering resistance
- Outstanding creep resistance

Customer: Kemflo

OEM/ End User: RIVIAN



AD-TCS Key Project

Roof Rack Base / ZHER

Application	Material	Customer (OEM, End-user)	Project No. or Segment	Pot. [t/a]	SOP	Resp. [Sales/AD]
Roof Rack Base	Grivory GV XE 16101 Black 9839 (GVL-5H HP)	Kemflo (RIVIAN)	230-250812-2	21	Q2 2026	ZHER /FUVI
Advantages (<i>Technical Competition</i>)			Risks / Competitor Grade			
+ High stiffness at elevated temperature + Outstanding creep resistance + Superior energy absorption			- Processing <i>During the processing, there was a dosing issue. For the better processing, the barrel temperature was increased to 320-330 °C, the processing is poor.</i>			
Actual Situation			Next Steps (<i>by When</i>)			Resp.
■ FEA done, GVL-5H passed with a factor of safety and the color was approved. ■ Mold trial done, functional test of these parts is on-going, the test result will be finished by 28th Feb. 2026.			■ Follow up the test result and Trial support (~28th .02.2026)			ZHER

AD-TCS Key Project

Lamellar Bar / ZHZA



Reference

Material:

Grinova S XE 10767 Black L11720.4 (XGF35)

Requirements

- High stiffness
- High surface glossiness
- Good scratch resistance

Customer: Chengdu AVIC

OEM / End User: Dongfeng

AD-TCS Key Project

Lamellar Bar / ZHZA

Application	Material	Customer <i>(OEM, End-user)</i>	Project No. or Segment	Pot. <i>[t/a]</i>	SOP	Resp. <i>[Sales/AD]</i>
Lamellar Bar	Grinova S XE 10767 Black L11720.4	Chengdu AVIC <i>(Dongfeng)</i>	230-260104-1	19	Q3.2026	XURO / ZHZA
Advantages <i>(Technical Competition)</i>			Risks / Competitor Grade			
+ Combines high structural stiffness and high surface glossiness + Technical support <i>(molding trial)</i>			- High material price			
Actual Situation			Next Steps <i>(by When)</i>			Resp.
- Customer approved the surface quality of sample produced with Grinova S - Products design ongoing			Support for customer tooling <i>(Q1 2026)</i>			ZHZA

AD-TCS Key Project

Lamellar Bar / ZHZA

Combined design,

High gloss surface + Matte finished surface



Hard Floor Panel under Console / SEOJ

Material:

Grivory GVX-5H Black 9915



Requirements

- High stiffness
- Scratch resistance
- Warpage below of 1mm(part length 1200mm)

Customer: KBI Dongkook Industry

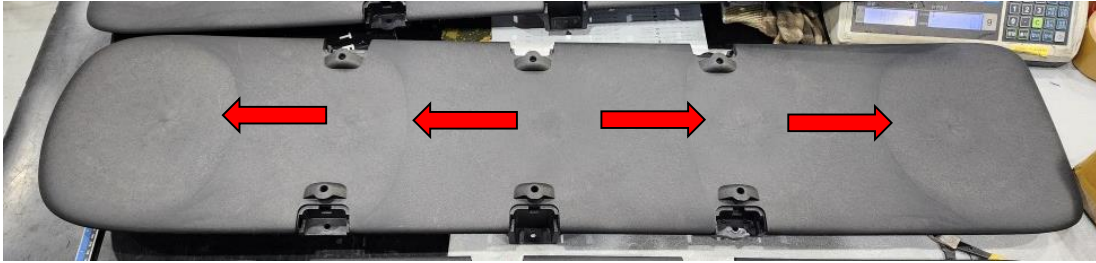
OEM/ End User: HKMC

Hard Floor Panel under Console / SEOJ

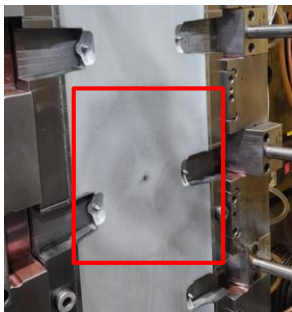
Application	Material	Customer (OEM, End-user)	Project No. or Segment	Pot. [t/a]	SOP	Resp. [Sales/AD]
Hard Floor Panel under Console	Grivory GVX- 5H Black 9915	KBI Dongkook Ind. (HKMC)	230-251114-1	60	Q1 2029	PAIS /SEOJ
Advantages (<i>Technical Competition</i>)			Risks / Competitor Grade			
+ High stiffness + Scratch resistance + Warpage below of 1mm (<i>part length 1,200mm</i>)			- Competitor BASF (PA6+X GF40%) vs. EMS GVX-5H <i>Density BASF : 1.46 vs EMS 1.56</i> - Warpage must be below 1mm <i>Part length 1,200 mm, Hot runner 5 points of sequential pinpoint gate system using and gate Ø 1.2mm.</i>			
Actual Situation			Next Steps (<i>by When</i>)			Resp.
■ Mold trial done, GVX showed less warpage compared with BASF in eyes. ■ Part surface needs improvement. A additional mold trial was scheduled. Recommendations for Tooling / Gating given by ADC. ■ Customer is producing Jig to measure warpage value.			■ Trial support to optimize surface quality (~30.01.26') ■ Check warpage measurement (~30.03.26')			SEOJ SEOJ

Hard Floor Panel under Console / SEOJ

Using sequential gate system



Erased make in tool



Improved surface



Overview List I&C

Application	Product	Customer (OEM / End-user)	Seg.	Pot. [t/a]	SOP	Resp. (Sales/AD)
Cow Collar Base Buckles	Grivory GVN-5H	ChangXin (Halter NZ)	720	160	03.2026	SUN WONJ
Drone Diverse Parts	Grivory G VX-5H	Cretia (Google)	760	150	06.2026	JIMJ LINR
E-scooter Display Housing	Grivory GV XE 11119 (GV-GFX50 V0)	Hualong (YADEA)	740	100	03.2026	MOSK ZHVI
HVAC Ball Valve	Grivory HT1V-5 FWA	JLT/Chuandibai (Grandsire)	410	50	06.2026	WEIC JIAJ
AGV Chassis	Grivory GV XE 16141 (GVCL-4H V0)	Chen'nuo (Geek+)	450	44	06.2026	GEJA ZHUP

Cow Collar Buckles / Halter NZ



Material:
Grivory GVN-5H

Requirements

- Pull out test >200 kgf after moisture conditioning
- IK08 Impact testing
- Effluent resistance

Customer: ChangXin

OEM/ End User: Halter NZ

Cow Collar Base Buckles / Halter NZ

Application	Material	Customer (OEM, End-user)	Project No. or Segment	Pot. [t/a]	SOP	Resp. [Sales/AD]
Cow Collar Base Buckles	GVN-5H	ChangXin (Halter NZ)	720-251118-1	160	02.2026	SUN WONJ
Advantages (<i>Technical Competition</i>)			Risks / Competitor Grade			
+ Reliably performance after moisture absorption + Great support from EMS			- n/a			
Actual Situation			Next Steps (<i>by When</i>)			Resp.
■ GVN-5H was suggested due to good performance after conditioning, 05.2025 ■ Price accepted and start tooling, 11.2025 ■ GVN-5H tooling ready and sampling done, 12.2025 ■ All test passed. 3t order received, 01.2026			■ Secure monthly order, monthly			SUN

Divers Drone Parts / Cretia



Material:

Grivory GVX-5H black

Requirements

- Cantilever beam 90° pendulum impact
- 1 meter, 100 grams, steel ball impact
- Thermal shock, -20~70°C, 72 hrs, yellowing DE≤1.0.

Customer: Cretia

OEM / End User: FIH / Google

AD-TCS Key Project

Divers Drone Parts / Cretia

Application	Material	Customer (OEM, End-user)	Project No. or Segment	Pot. [t/a]	SOP	Resp. [Sales/AD]
Divers Drone Parts	GVX-5H	Cretia (Google)	760-260109-1	150	06.2026	JIMJ LINR
Advantages (Technical Competition)			Risks / Competitor Grade			
+ Excellent low warpage + Robust mechanical properties to withstand impact test			- n/a			
Actual Situation			Next Steps (by When)			Resp.
■ Trial finished, prelim test passed at Cretia, 12.2025 ■ 5'000 sets are planning in Q1 2026, 01.2026			■ Support 5'000 sets pilot run, 28 Feb. 2026			LINR

E-scooter Display Housing / YADEA



Material:

Grivory GV XE 11119 (GV-GFX50 V0)

Requirements

- Low warpage than PC-GF, $<0.5\text{mm}$
- Thermal shock, -40 to 85°C , 75mins, 300cycles

Molder: Hualong

Tier: AISPEECH

OEM / End User: YADEA

E-scooter Display Housing / YADEA

Application	Material	Customer (OEM, End-user)	Project No. or Segment	Pot. [t/a]	SOP	Resp. [Sales/AD]
E-scooter Display Housing	XE 11119 (GV-GFX50 V0)	Hualong (YADEA)	740-251212-1	100	03.2026	MOSK / ZHVI
Advantages (<i>Technical Competition</i>)			Risks / Competitor Grade			
+ Low warpage + Good mechanical strength			- Possible local competitors involved after production			
Actual Situation			Next Steps (<i>by When</i>)			Resp.
▪ AI replacement and problem-solving of PC/ABS due to waterproof failure and low heat resistance, 11.2025 ▪ Trial finished, warpage was improved to <0.5mm, 200kg was delivered, 12.2025 ▪ 2t was delivered for pilot run, 01.2026			▪ Obtain test result and support pilot run, 02.2026			ZHVI

AD-TCS Key Project

Ball Valve / Grandsire

Material:
Grivory HT1V-5 FWA

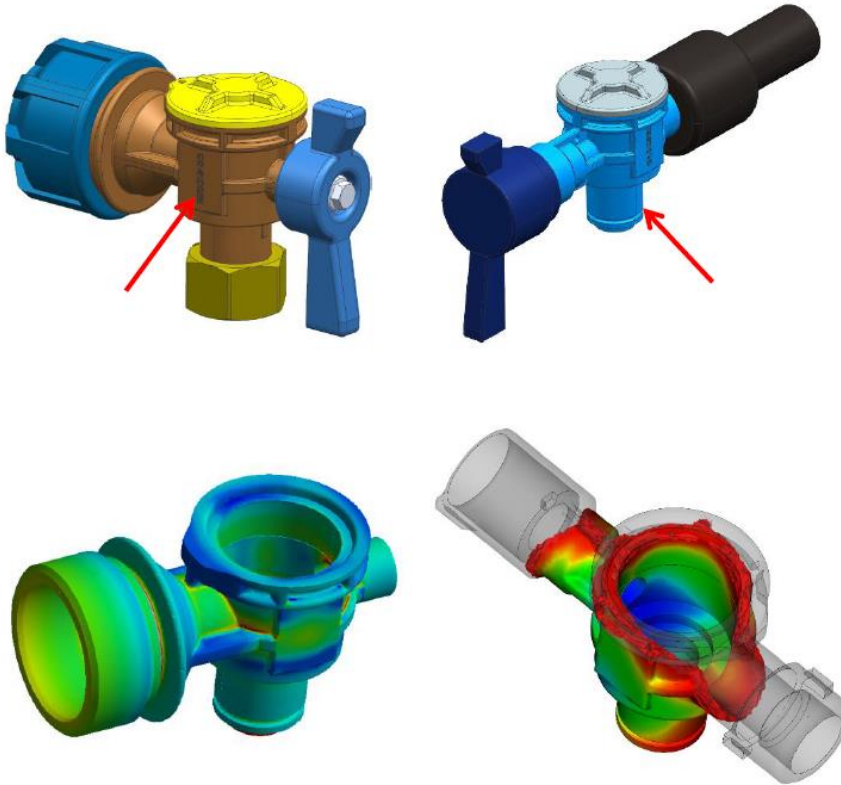
Requirements

- **High stiffness & strength**
(burst pressure > 60bar @23°C)
- **Food & water contact approved**

Customer: JLT/ Chuandibai

Tier: Sansheng

End User: Grandsire



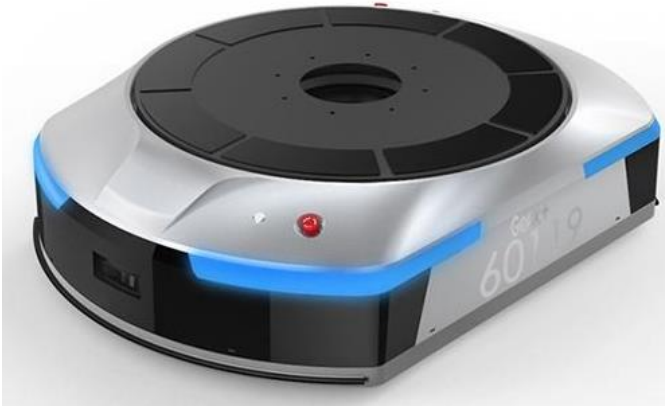
AD-TCS Key Project

Ball Valve / Grandsire

Application	Material	Customer (OEM, End-user)	Project No. or Segment	Pot. [t/a]	SOP	Resp. [Sales/AD]
Ball valve	Grivory HT1V-5 FWA	JLT/Chuandibai (Grandsire)	410-241011-1	50	06.2026	WEIC JIAJ, XUER
Advantages (Technical Competition)			Risks / Competitor Grade			
+ Technical support + High strength & stiffness			- n/a			
Actual Situation			Next Steps (by When)			Resp.
- Chuandibai (1st molder) part passed test and mass production started (9t material delivered), 11.2025 - JLT (2nd molder) parts passed all test, 01.2026 - Around 19 ton material were delivered, 01.2026			Track mass production status, 02.2026			JIAJ

AD-TCS Key Project

AGV Chassis / Geek+



Material:

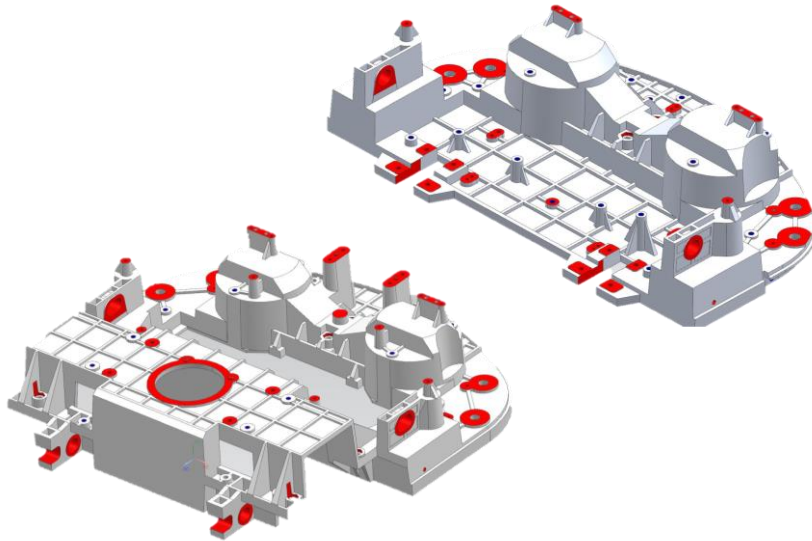
Grivory GV XE 16141 (GVCL-4H V0)

Requirements

- 600,000 cycles lifting test @ 40kg
- 150,000 times 50mm height obstacle tests
- 100 times collisions @ 4m/s
- Chassis deflection < 3mm @ 40kg

Customer: Chennuo

End User: Geek+



AD-TCS Key Project

AGV Chassis / Geek+

Application	Material	Customer (OEM, End-user)	Project No. or Segment	Pot. [t/a]	SOP	Resp. [Sales/AD]
AGV Chassis	XE 16141 (GVCL-4H V0)	Chennuo (Geek+)	450-250925-1	44	06.2026	GEJA ZHUP
Advantages (Technical Competition)			Risks / Competitor Grade			
+ Material performance (UL V0 & ESD) + Technical support (FEA, MFA, on-site trial)			- Changing requirement & situation of cost saving			
Actual Situation			Next Steps (by When)			Resp.
▪ New design AGV chassis, plastic solution has ~15% cost saving vs metal die-casting, 10.2025 ▪ FEA result showed acceptable deformation (1.5 vs. spec 2.0mm) & stress level (Max. 59 MPa), 10.2025 ▪ FEA for limit load condition (3,000 N) supported and design optimization implemented, 12.2025 ▪ Tooling was ready for trial, 01.2026			▪ Support sampling trial, 02.2026			ZHUP

Overview List CATS

Application	Product	Customer (OEM / End-user)	Seg.	Pot. [t/a]	SOP	Resp. (Sales/AD)
Thermostat Housing	Grivory HT1VA-35 HYS	Temb (JMC)	222	36	Sep. 2027	WANH/LIPE, WABR
Air Vent Trim	Grilamid TR XE 11859	JoysonQuin (FAW VW)	230	50	Dec. 2026	SONG/CHFI, JJO
Gear Fork	Grinova H XE 11627	DFFB (GWM)	270	15	Dec. 2026	WANH/XUJR, XUER
Hybrid CCB	Grilon BG-60 FC	Boyi (BAIC)	230	300	Dec. 2026	WAHU/HULO, ZHAN
TV Stand	Grivory GV-6H	Haier (Haier)	420	30	Dec. 2026	TIAH/ZHUP, LIDI

Thermostat Housing / Temb

Material:

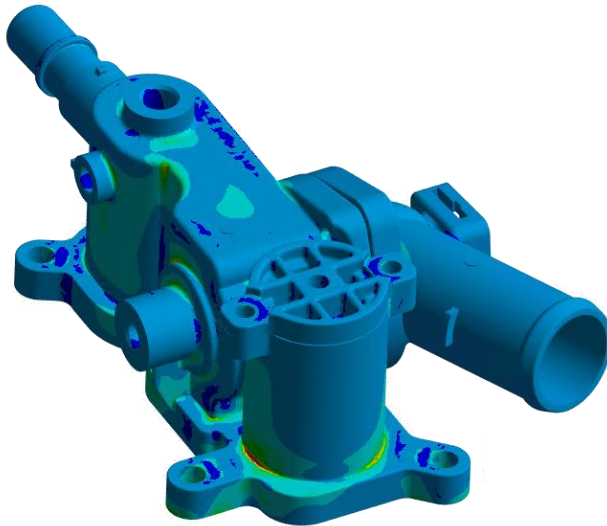
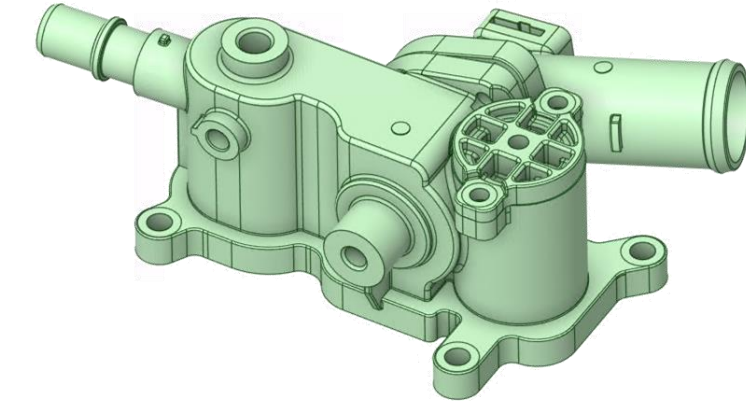
Grivory HT1VA-35 HYS

Requirements

- High strength & stiffness at high temperature (*inner pressure of 2.6bar & 1'900N at 120 °C, 2'000h*)
- High temperature resistance

Customer: Temb

OEM/ End User: JMC



AD-TCS Key Project

Thermostat Housing / Temb

Application	Material	Customer (OEM, End-user)	Project No. or Segment	Pot. [t/a]	SOP	Resp. [Sales/AD]
Thermostat Housing	Grivory HT1VA-35 HYS	Temb (JMC)	222-260107-2	36	Sep. 2027	WANH/LIPE, WABR
Advantages <i>(Technical Competition)</i>			Risks / Competitor Grade			
+ High strength/stiffness at high temperature + Success story + FEA support			- No			
Actual Situation			Next Steps <i>(by When)</i>			Resp.
- Design optimized based on FEA to improve structure strength. - FEA report provided to customer.			Follow up tooling trial by Jun. 2026			WABR

AD-TCS Key Project

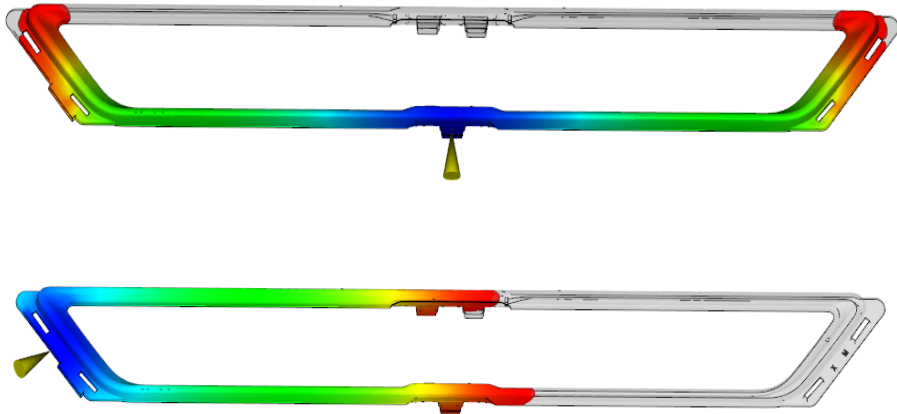
Air Vent Trim / FAW VW



Material:
Grilamid TR XE 11859

Requirements

- High gloss appearance
- Scratch resistance
- Cost saving compared to PC/ABC painting



Customer: JoysonQuin

OEM/ End User: FAW VW

AD-TCS Key Project

Air Vent Trim / FAW VW

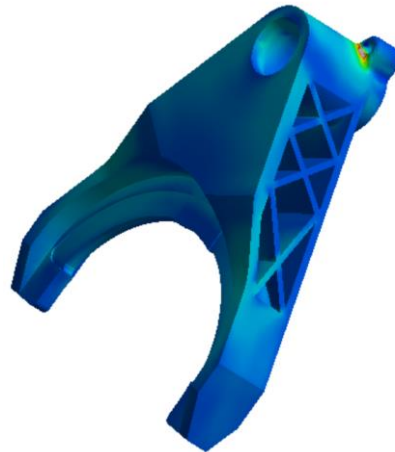
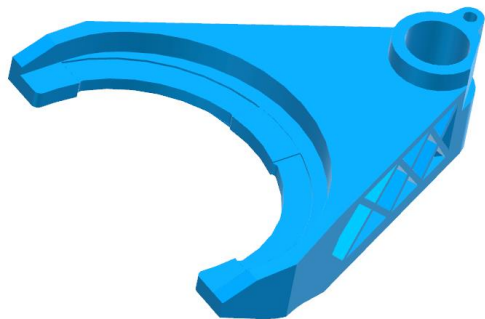
Application	Material	Customer (OEM, End-user)	Project No. or Segment	Pot. [t/a]	SOP	Resp. [Sales/AD]
Air Vent Trim	Grilamid TR XE 11859	FAW VW (JoysonQuin)	230-250227-1	50	Mar. 2026	SONG/CHFI, JJO
Advantages <i>(Technical Competition)</i>			Risks / Competitor Grade			
+ Cost saving compared to PC/ABS painting + MFA support			- PC/ABS with UV & piano black painting			
Actual Situation			Next Steps <i>(by When)</i>			Resp.
- MFA with two gating scenarios investigated to improve filling/warpage - MFA report provided to customer.			Follow up tooling by Mar. 2026			JJO

AD-TCS Key Project

Gear Fork / DFFB



Metal design



Optimized plastic design

Material:
Grinova H XE 11627 (GF60)

Requirements

- High stiffness at high temperature
(*deformation < 0.4mm at 200N @ 150 °C*)
- Wearing resistance
- Cost and weight saving

Customer: DFFB

OEM/ End User: GWM

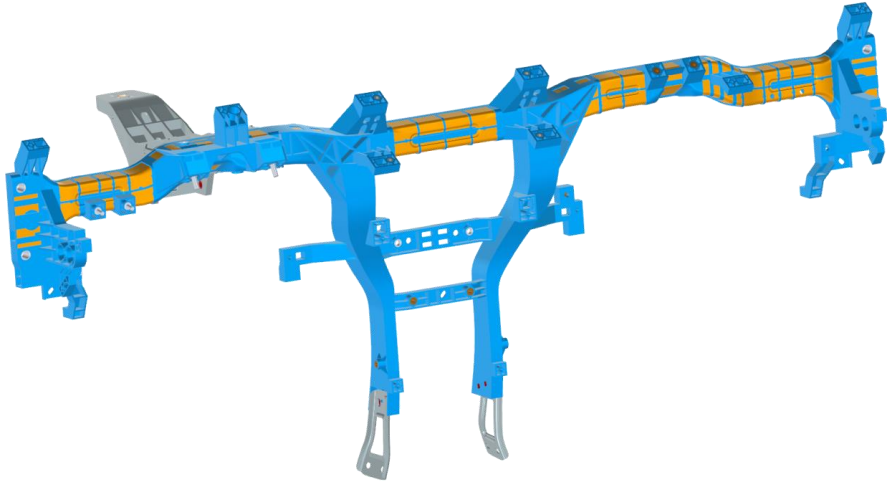
AD-TCS Key Project

Gear Fork / DFFB

Application	Material	Customer (OEM, End-user)	Project No. or Segment	Pot. [t/a]	SOP	Resp. [Sales/AD]
Gear Fork	Grinova H XE 11627	DFFB (GWM)	270-251226-1	15	Dec. 2026	WANH/XUJR, XUER
Advantages <i>(Technical Competition)</i>			Risks / Competitor Grade			
+ High stiffness at high temperature + Cost and weight saving + Design & FEA support			- Deformation			
Actual Situation			Next Steps <i>(by When)</i>			Resp.
■ Several rounds of design optimization carried out and validated with FEA, as DFFB can't provide a clear design space and often changing ■ All findings shared with DFFB			■ Follow up prototyping by Feb. 2026			XUER

AD-TCS Key Project

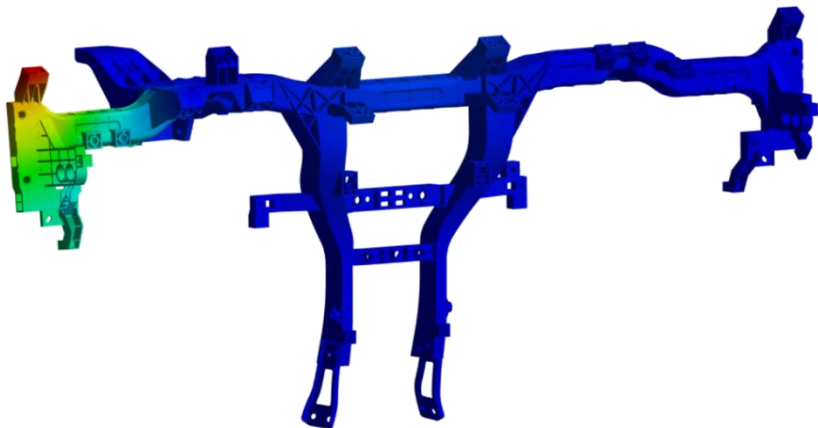
Hybrid CCB / Boyi



Material:
Grilon BG-60 FC

Requirements

- **High strength & stiffness**
(lateral test force of 15KN)
- **High flow ability**



Customer: Boyi

OEM/ End User: BAIC

AD-TCS Key Project

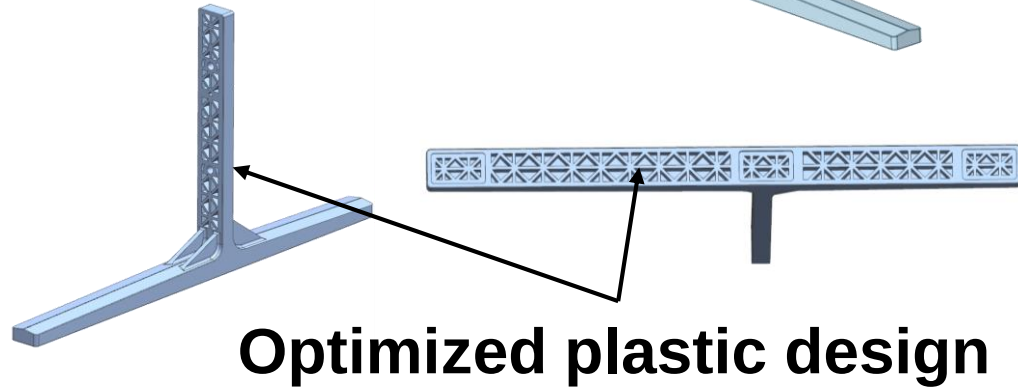
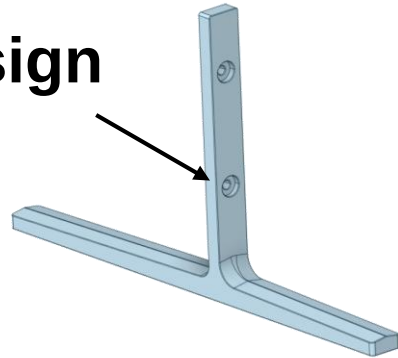
Hybrid CCB / Boyi

Application	Material	Customer (OEM, End-user)	Project No. or Segment	Pot. [t/a]	SOP	Resp. [Sales/AD]
Hybrid CCB	Grilon BG-60 FC	Boyi (BAIC)	230-251225-1	300	Dec. 2026	WAHU/HULO, ZHAN
Advantages <i>(Technical Competition)</i>			Risks / Competitor Grade			
+ High strength & stiffness + Success story + FEA support			- Local PA6 GF60			
Actual Situation			Next Steps <i>(by When)</i>			Resp.
■ FEA is ongoing.			■ Complete FEA by the end of Jan. 2026			ZHAN

AD-TCS Key Project

TV Stand / Haier

Aluminum design



Optimized plastic design

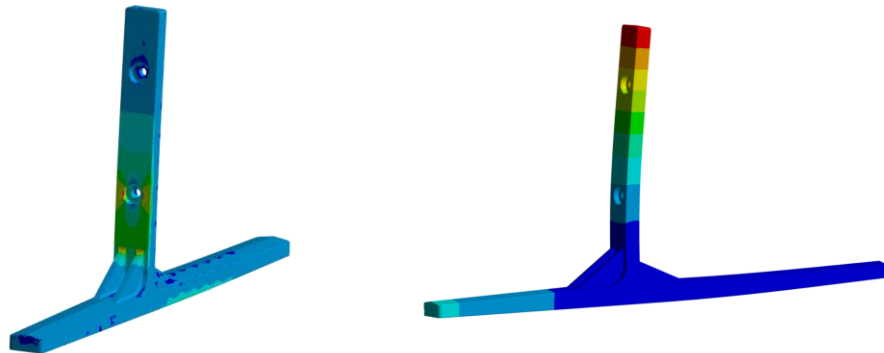
Material:
Grivory GV-6H

Requirements

- **High strength & stiffness**
(test load of 113Kg @ 23 °C)
- **Cost & weight reduction**

Customer: Haier

OEM/ End User: Haier



AD-TCS Key Project

TV Stand / Haier

Application	Material	Customer (OEM, End-user)	Project No. or Segment	Pot. [t/a]	SOP	Resp. [Sales/AD]
TV Stand	Grivory GV-6H	Haier (Haier)	AP 420-251127	30	Dec. 2026	TIAH/ZHUP, LIDI
Advantages (Technical Competition)			Risks / Competitor Grade			
+ High strength & stiffness + Cost saving + Design & FEA support			- Creep			
Actual Situation			Next Steps (by When)			Resp.
- Plastic oriented design was made and validated with FEA. - Report provided to customer for review.			Follow up customer's decision and support MFA for tooling as needed.			LIDI